

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool : Case Study

Weightage: 10%

QUESTIONS:

Total Marks: 10

Scenario: Web Development Standards Implementation

A startup is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

- A. `<media src="video.mp4"></media>`
- B. `<video src="video.mp4" controls></video>`
- C. `<movie src="video.mp4"></movie>`
- D. `<vid src="video.mp4"></vid>`

Snippet 2: Form Submission Methods

- A. GET
- B. POST

- C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. <div>
- B. <nav>
- C. <section>
- D. <header>

Snippet 4: XML Syntax

- A. <note><to>User</to><from>Admin</from>
- B. <note><to>User</to><from>Admin</from></note>
- C. <note><to>User</from></note>
- D. <note to="User"></note>

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Questions:

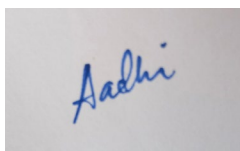
1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

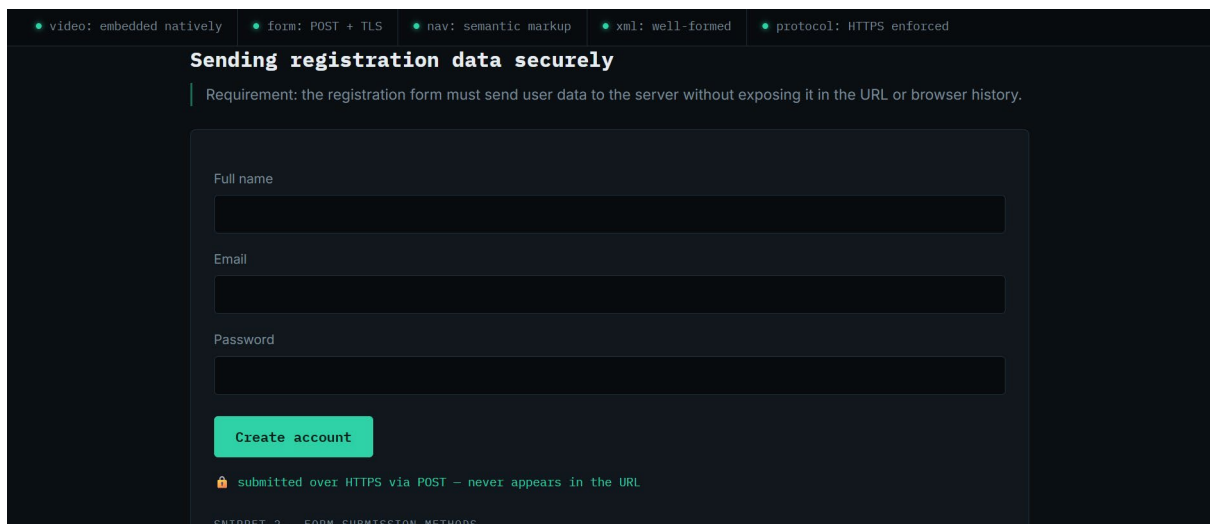
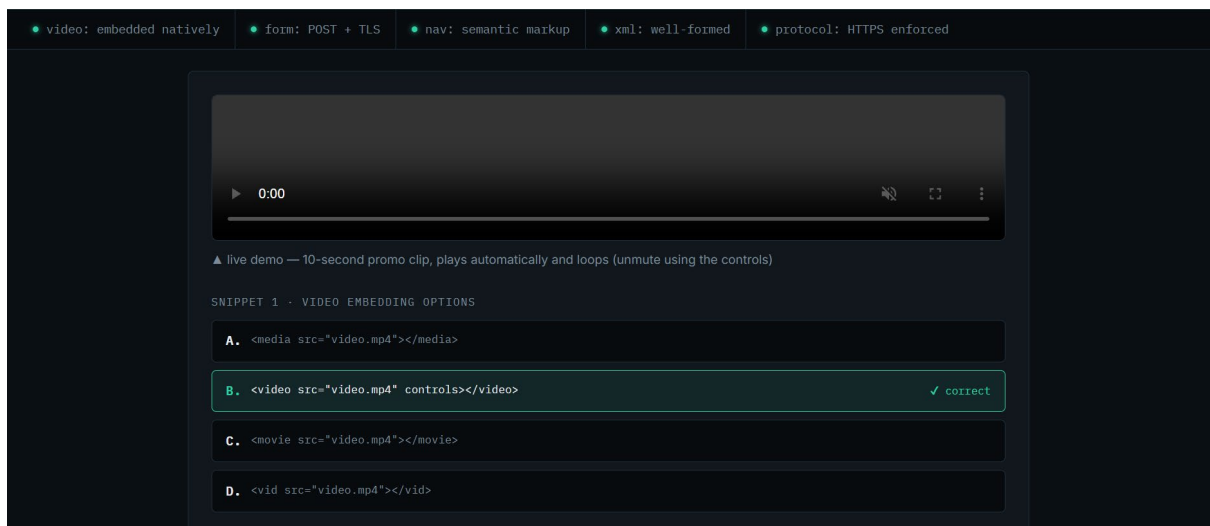
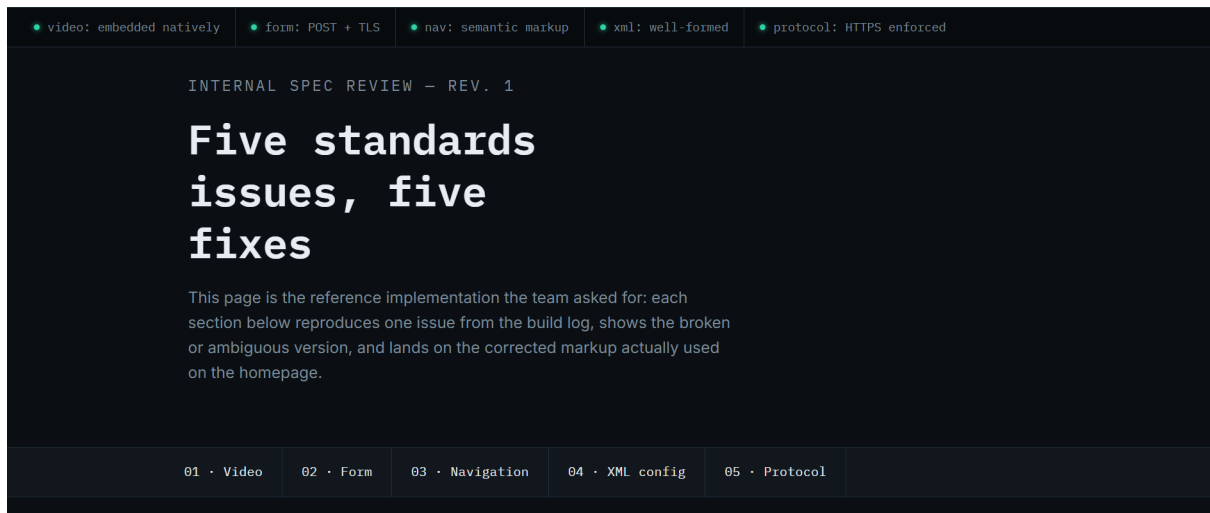
Code of Conduct:

I Aadhirai S P (192411002) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in blue ink that reads "Aadhi".



video: embedded natively

form: POST + TLS

nav: semantic markup

xml: well-formed

protocol: HTTPS enforced

🔒 submitted over HTTPS via POST – never appears in the URL

SNIPPET 2 · FORM SUBMISSION METHODS

A. GET

B. POST ✓ correct

C. PUT

D. DELETE

Q2 · IDENTIFY THE APPROPRIATE HTTP METHOD FOR FORM SUBMISSION AND EXPLAIN WHY.

Answer: B — POST. POST sends the form fields in the request body rather than the URL, which keeps sensitive data like passwords out of the address bar, browser history, and server access logs. GET (A) exposes data as a visible query string, which is fine for a search box but wrong for a registration form. PUT (C) and DELETE (D) are REST methods meant for updating or removing an existing resource at a known URL — a new registration is a resource being created, and browser <form> elements don't natively support PUT/DELETE anyway.

video: embedded natively

form: POST + TLS

nav: semantic markup

xml: well-formed

protocol: HTTPS enforced

BEFORE — DIV SOUP

<div class="menu"> <div class="menu-item">Home</div> <div class="menu-item">Pricing</div> </div>

AFTER — SEMANTIC MARKUP

<nav aria-label="Primary"> Home Pricing </nav>

SNIPPET 3 · NAVIGATION ELEMENTS

A. <div>

B. <nav> ✓ correct

C. <section>

D. <header>

video: embedded natively

form: POST + TLS

nav: semantic markup

xml: well-formed

protocol: HTTPS enforced

BEFORE — THREE SYNTAX ERRORS

<?xml version="1.0"?> <config> <database> <host>localhost<host> <!-- 1: closing tag should be </host> --> <port>5432</port> <!-- 2: missing ">" after <port --> </database> <timeout>30</timeout> <!-- 3: tag names are case-sensitive --> <config> <!-- 4: missing "/" – root never closes -->

AFTER — WELL-FORMED

<?xml version="1.0" encoding="UTF-8"?> <config> <database> <host>localhost</host> <port>5432</port> </database> <timeout>30</timeout> </config>

SNIPPET 4 · XML SYNTAX

A. <note><to>User</to><from>Admin</from>

B. <note><to>User</to><from>Admin</from></note> ✓ correct

C. <note><to>User</from></note>

D. <note to="User"></note>

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand